

Dilawar Singh

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Founding / Senior Systems Engineer with 10+ years of experience building kernel-level components (eBPF, Windows minifilters), endpoint and data security systems, SaaS platforms, and real-time simulators. Proven record of taking deep-tech products from PoC to production: designing core architectures, writing performance-critical low-level code in Rust and C/C++, and shipping cross-platform systems. Enthusiastic contributor to Free and Open Source Software (FOSS).

Skills

- **Systems & Low-Level:** Linux kernel modules, eBPF (Extended Berkeley Packet Filter), Windows Kernel Minifilter (WDK), endpoint security, real-time systems, Win32/POSIX APIs, IPC, SDK development (Rust, C++), GUI (Qt, ImGui), CLI tooling
- **Languages:** Rust, C, C++, Python (expert); Java, TypeScript, JavaScript, Nim, Haskell (intermediate); VHDL, Verilog, SystemC, BlueSpec (familiar); PHP
- **Backend & SaaS:** Microservices and distributed systems; high-performance gRPC/REST APIs (Axum, Actix, FastAPI); PostgreSQL, DuckDB; job queues; event-driven systems; data parsing, validation, and lifecycle management
- **Observability & Infrastructure:** OpenTelemetry, Grafana; Docker, QEMU; CI/CD (GitLab, Jenkins); build systems; packaging (OBS, NSIS); Ansible, Pulumi
- **Scientific & Computation:** Boost, GNU Scientific Library (GSL), SciPy stack, OpenCV; numerical simulators; pattern matching and optimization (network flows, graphs, linear algebra); stochastic DSA; parallelism partitioning
- **Other:** Systems architecture, parser development, technical literature review, team leadership and hiring

Work Experience

Independent Consultant — Self-Employed, Bengaluru Mar 2026 – Present

- Designing and developing scalable microservices supporting a virtual assistant platform for an automotive control system (Rust, C, Qualcomm SoC)
- Designing and overseeing development of high-performance APIs (Rust, AMQP) for a cyber lab running near-real-time experiments on highly correlated, diverse datasets (audio, video, sensor events)

Senior Engineer — Dognosis, Bengaluru Jan 2025 – Feb 2026

- Co-developed high-performance async REST APIs in Rust (Poem, Tokio) powering the sensor data ingestion and query layer
- Aggregated ~30 GB/session and ~10,000 events/session of live sensor data into a single operational control plane used for real-time decision-making and incident response
- Set up and operated self-hosted infrastructure (GitLab, PostgreSQL, Grafana, CI/CD runners), including access control, backups, and deployment policies
- Established risk-focused planning and sprint rituals that reduced invisible work, clarified ownership, and gave leadership visibility into team throughput

- Built hiring pipelines and hired 5 engineers, scaling the team from 6 to 11 in 3 months

Cofounder & CTO — Subconscious Compute, Bengaluru Dec 2019 – Dec 2024

- Wrote Windows kernel minifilter (WDK, C++/Rust) and Linux eBPF sensors generating real-time telemetry for Shepherd's cross-platform threat detection and compliance engine, deployed across customer endpoints
- Authored the Windows port of Shepherd in Rust, including unsafe Win32 API bindings and FFI interop with existing C++ components
- Hired and led a team of 15 systems engineers to build Shepherd, an automated endpoint compliance and security platform
- Authored technical memos and pitch decks to secure accelerator placements and support fundraising

Research Fellow & Google Summer of Code (GSoC) Mentor 2016 – 2018

- Added high-performance numerical solvers (C++, Boost, GNU Scientific Library) to the [MOOSE simulator](#), including a [standalone LSODA solver](#), enabling it to solve multi-scale neural simulation problems
- Wrote Python bindings (pybind11) and created packages for [MOOSE](#) and [Smoldyn](#), significantly improving usability and adoption by the scientific community
- Mentored multiple Google Summer of Code (GSoC) contributors on GPU/CUDA-based acceleration over 4 years

Firmware Engineer — Kritikal Solutions, Noida Jul 2009 – Jun 2010

- Implemented a Kalman-filter-based image stabilization system for professional movie cameras (C++, multi-threaded RTOS) to eliminate vibration artifacts from recordings

Projects

- Published Rust crates: [simple_accumulator](#) (online statistics), [query-wmi](#) (Windows WMI interface), [job-dispatcher](#) (async job execution)
- Published Python packages: [PlotDigitizer](#) (extract raw data from plot images), [SerialScope](#) (serial-port oscilloscope), [eye-blink-detector](#) (OpenCV-based computer vision)

Education

- **PhD, Computational Neuroscience** — NCBS Bangalore (TIFR Mumbai), 2019
- **PhD, Digital Systems** (withdrawn) — IIT Bombay, 2011–2013
- **M.Tech, Electrical Engineering (VLSI)** — IIT Bombay, 2009
- **B.Tech, Instrumentation & Control Engineering** — Dr. MGR ERI Chennai, 2007

Publications

[Full list on Google Scholar](#)

- [Subunit exchange enhances information retention by CaMKII in dendritic spines](#), *eLife*, 2018
- [Python interfaces for the Smoldyn simulator](#), *Bioinformatics*, 2022
- [BioSimulators: a central registry of simulation engines](#), *Nucleic Acids Research (NAR)*, 2022